

BRENDAN HENRIQUE

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EDUCATION

- 2021–Present **University of California, Berkeley**
Ph.D. Candidate, Learning Sciences and Human Development
Berkeley School of Education
Expected completion: May 2027
Dissertation: *Ideologies, Motivations, and Identities in Critical CS Teacher Education*
Committee: Michelle Wilkerson (Chair), Thomas Philip, Lisa Yan
- 2021 **Loyola Marymount University**
Master of Arts in Urban Education
Advisor: Prof. Maryann Krikorian
- 2019 **University of California, Berkeley**
Bachelor of Arts in Cognitive Science, High Honors
Thesis Advisors: Prof. Michael Ranney and Prof. Glynda Hull

RESEARCH APPOINTMENTS

- Graduate Student Researcher** August 2025 – Present
Department of Electrical Engineering and Computer Sciences, UC Berkeley *Berkeley, CA*
PI: Prof. Lisa Yan
- Contribute to *CUE-T: A Partnership to Scale Inclusive Undergraduate Data Science through an Interdisciplinary, Introductory Course in Computing and Social Science* (NSF IUSE).
 - Lead a curricular review and cataloging study analyzing how disciplinary content and computing skills are integrated across multiple course iterations and institutions.
 - Supervise two undergraduate research assistants in data collection and qualitative analysis; both are co-authors on a peer-reviewed conference presentation.
- Graduate Student Researcher** January 2023 – August 2025
CORE Lab, Berkeley School of Education, UC Berkeley *Berkeley, CA*
PIs: Prof. Michelle Wilkerson (UC Berkeley) & Dr. Tamar Fuhrmann (Columbia University)
- Contributed to *From Access to Sustainability* (NSF), investigating how computational modeling practices are sustained in K–12 science classrooms.
 - Led a design team developing a middle school biology unit in which students use computer code to model mechanistic reasoning in photosynthesis.
 - Conducted qualitative analysis of classroom video and student artifacts; contributed to multiple peer-reviewed publications and presentations from this work.
- Undergraduate Researcher** Fall 2017 – Spring 2019
Hull Research Group, Berkeley School of Education, UC Berkeley *Berkeley, CA*
PI: Prof. Glynda Hull
- Analyzed middle school students' 360-degree virtual reality storytelling using a human-computer interaction framework. Developed and implemented a digital literacy curriculum on the Space2Cre8 social network, using qualitative methods to analyze video-mediated creative collaboration.

PEER-REVIEWED JOURNAL ARTICLES

[†] Undergraduate mentee co-author

1. Fuhrmann, T., Rosenbaum, L. F., Eloy, A., Wagh, A., **Henrique, B.**, Blikstein, P., & Wilkerson, M. H. (2026). Designing MoDa: A new tool for science learning through data-based computational modeling. *International Journal of Child-Computer Interaction*, 48, 100819. <https://doi.org/10.1016/j.ijcci.2026.100819>

Last updated: July 29, 2026

PEER-REVIEWED CONFERENCE PROCEEDINGS: PAPERS

1. Wagh, A., Rosenbaum, L. F., Eloy, A., Fuhrmann, T., **Henrique, B.**, Blikstein, P., & Wilkerson, M. H. (2025). Designing for discourse: Enacting responsive teaching through code reflections in a computational modeling unit [Long paper]. In A. Rajala, A. Cortez, R. Hofmann, A. Jornet, H. Lotz-Sisitka, & L. Markauskaite (Eds.), *Proceedings of the 19th International Conference of the Learning Sciences — ICLS 2025* (pp. 377–385). International Society of the Learning Sciences.
2. **Henrique, B.** (2025). Exploring critical CS teacher education program design through a science and technology studies approach [Full paper]. In *Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 1* (pp. 478–484). Association for Computing Machinery. <https://doi.org/10.1145/3641554.3701903>
3. **Henrique, B.** (2024). “An act of a revolution”: Critical learning trajectories in a pre-service teacher computer science course [Short paper]. In R. Lindgren, T. I. Asino, E. A. Kyza, C. K. Looi, D. T. Keifert, & E. Suárez (Eds.), *Proceedings of the 18th International Conference of the Learning Sciences — ICLS 2024* (pp. 1802–1805). International Society of the Learning Sciences. <https://doi.org/10.22318/icls2024.115107>

PEER-REVIEWED CONFERENCE PROCEEDINGS: POSTERS AND EXTENDED ABSTRACTS

1. **Henrique, B.**, Yan, L., Cano Low, F. G.[†], & Arriaga, F.[†] (2026, June). *IUSE-CUE: A collaboratively designed, interdisciplinary introduction to computational thinking in data science* [Poster, in press]. In *Proceedings of the American Society for Engineering Education (ASEE) Annual Conference & Exposition: NSF Grantee Poster Session*, Charlotte, NC, United States.
2. Fuhrmann, T., **Henrique, B.**, & Blikstein, P. (2026, June). *When code shapes scientific language: Speaking photosynthesis in code* [Poster, in press]. In *Proceedings of the International Society of the Learning Sciences (ISLS) Annual Meeting*, Irvine, CA, United States.
3. **Henrique, B.**, & Fischer, K.[†] (2026). A statewide analysis of ethics and social impact in the CS teacher preparation pipeline [Poster]. In *Proceedings of the 57th ACM Technical Symposium on Computer Science Education V. 2* (pp. 1355–1356). Association for Computing Machinery. <https://doi.org/10.1145/3770761.3777352>
4. **Henrique, B.** (2024). Exploring K–12 teachers’ trajectories of engagement with critically conscious computing [Poster, Graduate Student Research Competition]. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 2* (p. 1880). Association for Computing Machinery. <https://doi.org/10.1145/3626253.3635415> **First Place.**

PEER-REVIEWED CONFERENCE PRESENTATIONS

1. Fuhrmann, T., & **Henrique, B.** (2026, August). *From code to chloroplasts: Modeling as a language for mechanistic reasoning in science*. In *Mechanistic reasoning in biology: Exploring processes, challenges, and educational value* [Symposium]. Conference of European Researchers in Didactics of Biology (ERIDOB), Ljubljana, Slovenia.
2. Fuhrmann, T., **Henrique, B.**, Eloy, A., Rosenbaum, L. F., Wagh, A., Blikstein, P., & Wilkerson, M. H. (2026, April). *From code to chloroplasts: Speaking about photosynthesis through computational modeling* [Paper presentation]. National Association for Research in Science Teaching (NARST) Annual International Conference, Seattle, WA, United States.
3. **Henrique, B.** (2026, April). *Designated identity narratives: Teacher philosophy statements in critical CS education* [E-Lightning session]. American Educational Research Association (AERA) Annual Meeting, Los Angeles, CA, United States.
4. **Henrique, B.** (2024, April). *Understanding how elementary school pre-service teachers integrate CS concepts into their teaching practice* [Roundtable]. American Educational Research Association (AERA) Annual Meeting, Philadelphia, PA, United States.
5. Fuhrmann, T., & **Henrique, B.** (2023, October). *MoDa: Exploring computational modeling from a constructionist perspective* [Workshop]. Constructionism 2023, New York, NY, United States.

BOOK REVIEWS AND OTHER PUBLICATIONS

1. **Henrique, B.**, Roberto, C., & Wilkerson, M. H. (2023). Who creates our computational worlds? A review of *Critically Conscious Computing: Methods for Secondary Education*. *International Journal of Child-Computer Interaction*, 35, 100546. <https://doi.org/10.1016/j.ijcci.2022.100546>

FELLOWSHIPS AND GRANTS

2026	Special Projects Grant , ACM SIGCSE (\$5,000) <i>Sole investigator; competitively awarded for a qualitative interview study</i>
2021–2025	Pinto-Fialon Fellowship , UC Berkeley (\$50,000)
2017	Cal Teach Koshland Excellence-in-Teaching Scholarship , UC Berkeley (\$929)

TRAVEL AWARDS

2025	Barbara White Bequest Annual Meeting Travel Grant (\$1,550)
2024–2026	Campus Travel Grants (ASUC, Graduate Division), UC Berkeley (\$5,400)
2024–2026	Berkeley School of Education Travel Grant, UC Berkeley (\$900)

AWARDS AND HONORS

2026	Doctoral Consortium , International Computing Education Research (ICER) Competitive selection with travel support (\$500).
2024	First Place, Graduate Student Research Competition , ACM SIGCSE (\$500) First place in the graduate division of the ACM SIGCSE research competition.
2021	Graduate Library Research Award, Honorable Mention , Loyola Marymount University Recognized for excellence in master's thesis research and engagement with library collections.
2020	100 Teachers Award , West Contra Costa Unified School District Selected by administrators, families, and colleagues as one of 100 educators recognized from a district of 1,700 teachers.
2020	Outstanding Graduate Student Instructor (OGSI) Award , UC Berkeley Nominated by the Department of Electrical Engineering and Computer Sciences for teaching in 2019; received as an undergraduate student instructor.

TEACHING EXPERIENCE

Graduate Student Instructor <i>University of California, Berkeley</i>	Fall 2021 – Present <i>Berkeley, CA</i>
• EDUC 144: Practicum in Education (Fall 2023 – Present): Mentored over 150 undergraduate students through their capstone research course. Students conduct independent qualitative research projects from research design through written papers based on fieldwork at sites across the Bay Area. • EDUC 140: The Art of Making Meaning (Spring 2022 – Fall 2022, Summer 2024): Facilitated weekly discussions on theories of learning and literacy to support students as they learn qualitative research methods. • CS 10: The Beauty and Joy of Computing (Fall 2021): Led weekly lab and discussion sections for an introductory computing course using the block-based Snap! language.	
Middle School Science Teacher <i>West Contra Costa Unified School District</i>	June 2019 – June 2021 <i>Richmond, CA</i>
• Taught NGSS-aligned 7th and 8th grade science, incorporating Universal Design for Learning and culturally responsive pedagogy for 180+ students annually • Engaged students through project-based science, justice-centered STEM, and computational modeling.	
Undergraduate Student Instructor <i>CS 10: The Beauty and Joy of Computing, UC Berkeley</i>	Fall 2018 – Spring 2019 <i>Berkeley, CA</i>
• Co-developed and taught a weekly discussion section. • Collaborated with the Head TA to integrate equitable pedagogy training for undergraduate lab assistants.	

CURRICULUM DEVELOPMENT

EDUC 228B/229A: Python Programming for Teachers and CS Methods

Summer 2023

CS Authorization Sequence, Berkeley Teacher Education Program

Berkeley, CA

- Redesigned the program's culminating CS authorization courses, merging what had been separate programming and pedagogy offerings into an integrated sequence bridging computing content and teaching practice through Science and Technology Studies.
- Developed a seven-week curriculum in which teachers learn core CS concepts (functions, abstraction, sequences, data structures, and introductory data science) alongside critical texts, examining each computing skill in conversation with its social impact.

EDUC 224: Introduction to Block-Based Programming for Teachers

Summer 2022

CS Authorization Sequence, Berkeley Teacher Education Program

Berkeley, CA

- Redesigned an existing course to be taught entirely in Snap!, adapting the Beauty and Joy of Computing lab sequence for a teacher audience and structuring the course around three shifting roles (learner, teacher, and academic) so that teachers encountered each concept from all three positions.
- Paired programming content with critical pedagogical and computing education texts. Students participated in Snap!Con, a computer science education conference.

UNDERGRADUATE RESEARCH MENTORSHIP

All mentees are undergraduate researchers at UC Berkeley.

2025–Pres.	Kyle Fischer (URAP) Co-author, SIGCSE TS 2026
2025–Pres.	Fernanda Guadalupe Cano Low Co-author, ASEE 2026
2025–Pres.	Frida Arriaga (URAP) Co-author, ASEE 2026
2024–2025	Kayla Almero M.A. Student, UCLA Teacher Education Program
2023–2025	Erika Bruciaga M.A. Student in Psychology, San José State University
2023–2026	Ella Nikfar Psy.D. Student (entering Fall 2026)

SERVICE

Reviewing

2025–Pres.	Reviewer, ACM Technical Symposium on Computer Science Education (SIGCSE TS)
2025–Pres.	Reviewer, International Society of the Learning Sciences (ICLS/CSCL)
2025	Reviewer, ACM Conference on Human Factors in Computing Systems (CHI)
2025	Reviewer, Interaction Design and Children (IDC)
2022	Co-reviewer (with M. H. Wilkerson), <i>Journal of the Learning Sciences</i>

Conference Service

2024, 2025	Student Volunteer, ACM Technical Symposium on Computer Science Education
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Professional and Community Service

2020–Pres.	Treasurer and Board Member, Love.Learn.Success, Berkeley, CA Approve annual budgets and compensation, review and revise organizational mission, and advise executive leadership for an after-school nonprofit serving under-resourced communities across the East Bay.
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ADDITIONAL PROFESSIONAL EXPERIENCE

Mathematics Assessment Item Writer

January 2023 – Present

Achievement Network (ANet)

Boston, MA

- Developed and refined over 250 mathematics assessment items for grades 5–8, aligned to Common Core State Standards and reviewed for cognitive complexity and accessibility.

- Adapted artificial intelligence curricula for U.S. K–12 contexts and developed instructional materials for Snap! in collaboration with Jens Mönig; moderated the Snap! educator community.

MEDIA COVERAGE

1. Schwartz, S. (2021, November 23). COVID-19 is a science lesson waiting to happen. *Education Week*.
<https://www.edweek.org/teaching-learning/covid-19-is-a-science-lesson-waiting-to-happen/2021/11> (Featured teacher; profiled on designing a standards-aligned unit on COVID-19 vaccines responsive to community concerns.)
2. Johnson, S. (2020, December 15). One more challenge for California teachers in distance learning: All new science curriculum. *EdSource*.
<https://edsource.org/2020/one-more-challenge-for-teachers-in-distance-learning-all-new-science-curriculum/645101> (Featured teacher; profiled on adapting NGSS instruction and scientific argumentation for distance learning.)